

**Supplementary Materials for “Corticosterone Concentration Varies with Metabolic Rate in Vertebrates”**

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The following supplementary materials for “Corticosterone Concentration Varies with Metabolic Rate in Vertebrates” contain all data and associated references (Table S1), the phylogenetic tree of all reptiles, birds, and mammals included in analyses (Figure S1), BIC model selection criteria (Table S2), and plots showing the relationships between lifespan and corticosterone both before and after accounting for the effects of mass-specific metabolic rate (Figure S2).

**Table S1:** Data include body mass (Mass; g), mass-specific metabolic rate (MSMR; mW/g), lifespan (LS; years), baseline/elevated plasma corticosterone concentrations (CCST; ng/ml), assay type (RIA = radioimmunoassay, EIA = enzyme immunoassay, ELISA = enzyme-linked immunosorbent assay, DIDA = double isotope dilution assay), time to bleed (Time; min), and stressor type for reptiles, birds, and mammals used in this study. Sample size (N), sex (M = males, F = females, MF = both), season (B = breeding, NB = non-breeding), environment (C = captive, F = field), and baseline plasma cortisol concentrations (ng/ml) are also included, with “U” indicating information unspecified. Temperature (Temp; °C) is included for reptiles. Asterisks (\*) indicate serum glucocorticoid hormone concentrations and carrot symbols (^) indicate juveniles.

| Species                                        | Mass<br>(g) | MSMR<br>(mW/g) | LS<br>(yrs) | Temp<br>(°C) | N     | Sex | Season | Envr | Baseline<br>CCST<br>(ng/ml) | Elevated<br>CCST<br>(ng/ml) | Baseline<br>cortisol<br>(ng/ml) | Time<br>(min) | Stressor <sup>1</sup> | Assay                   |
|------------------------------------------------|-------------|----------------|-------------|--------------|-------|-----|--------|------|-----------------------------|-----------------------------|---------------------------------|---------------|-----------------------|-------------------------|
| <b>Reptiles</b>                                |             |                |             |              |       |     |        |      |                             |                             |                                 |               |                       |                         |
| <i>Agkistrodon piscivorus</i>                  | 421         | [1]            | 24.6        | [2]          | -     | 32  | MF     | U    | F                           | 8.77                        | 56.97                           | >3            | Restraint             | EIA [1]                 |
| <i>Alligator mississippiensis</i> <sup>^</sup> | 793         | [3]            | 0.14        | [4]          | 85.0  | [2] | -      | 10   | U                           | U                           | C                               | 2.46          | 17.35                 | Restraint RIA [3]       |
| <i>Amblyrhynchus cristatus</i>                 | 2227        | [4]            | 28.0        | [2]          | 20.5  | 98  | U      | U    | F                           | 4.61                        | 11.02                           | <3            | Restraint             | RIA [5]                 |
| <i>Anolis carolinensis</i>                     | 3           | [6]            | 0.95        | [7]          | 11.0  | [2] | 20.0   | 42   | F                           | U                           | C                               | 12.83         | -                     | <3 - RIA [6]            |
| <i>Anolis sagrei</i>                           | 5           | [8]            | 2.20        | [9]          | 8.0   | [2] | -      | 114  | M                           | U                           | F                               | 9.86          | -                     | >3 - RIA [8]            |
| <i>Antaresia childreni</i>                     | 384         | [4]            | 0.51        | [7]          | 30.0  | [2] | 29.6   | 16   | F                           | NB                          | C                               | 6.02          | 14.78                 | Restraint RIA [10]      |
| <i>Aspidoscelis sexlineatus</i>                | 8           | [4]            |             |              | 32.0  | 35  | MF     | U    | C                           | 13.13                       | 41.85                           | >3            | Restraint             | RIA [11]                |
| <i>Aspidoscelis uniparens</i>                  | 7           | [12]           |             |              | 32.0  | 17  | U      | U    | C                           | 8.04                        | 17.26                           | >3            | Restraint             | RIA [12]                |
| <i>Boa constrictor</i>                         | 3950        | [13]           | 0.10        | [4]          | 40.4  | [2] | -      | 15   | MF                          | U                           | C                               | 3.31          | -                     | - EIA [13]              |
| <i>Boaedon fuliginosus</i>                     | 182         | [14]           | 0.21        | [15]         | 27.0  | 24  | MF     | U    | C                           | 72.44                       | 79.43                           | >3            | Other                 | ELISA [14]              |
| <i>Caiman latirostris</i> <sup>^</sup>         | 161         | [16]           |             |              | 22.0  | [2] | 31.4   | 24   | U                           | U                           | C                               | 22.50         | -                     | - RIA [16]              |
| <i>Caretta caretta</i>                         | 156500      | [4]            |             |              | 76.0  | [2] | 28.1   | 46   | U                           | U                           | F                               | 1.19          | 7.59                  | Restraint RIA [17]      |
| <i>Chelonia mydas</i>                          | 160000      | [4]            | 0.07        | [18]         | 75.0  | [2] | -      | 20   | MF                          | U                           | F                               | 2.04          | -                     | 1.78 - ELISA [19]       |
| <i>Chelonoidis niger</i>                       | 118535      | [20]           |             |              | 177.0 | [2] | -      | 16   | MF                          | U                           | C                               | 0.81          | -                     | >3 - RIA [21]           |
| <i>Crocodylus johnstoni</i>                    | 141832      | [22]           | 0.13        | [23]         | 20.0  | [2] | -      | 44   | MF                          | U                           | F                               | 4.02          | 40.02                 | >3 Restraint RIA [22]   |
| <i>Crocodylus niloticus</i> <sup>^</sup>       | 540         | [24]           |             |              | 80.0  | [2] | 30.0   | 5    | U                           | U                           | C                               | 5.00          | 11.10                 | Other RIA [24]          |
| <i>Crotalus atrox</i>                          | 480         | [25]           |             |              | 27.0  | [2] | -      | 12   | M                           | U                           | C                               | 32.87         | 140.00                | <3 Restraint RIA [25]   |
| <i>Crotalus viridis</i>                        | 800         | [26]           |             |              | 31.0  | [2] | -      | 30   | M                           | U                           | F                               | 28.44         | 39.01                 | >3 Other RIA [26]       |
| <i>Elgaria coerulea</i>                        | 20          | [27]           |             |              | 8.0   | [2] | 23.5   | 15   | MF                          | U                           | C                               | 12.77         | 65.46                 | <3 ACTH RIA [27]        |
| <i>Elgaria multicarinata</i>                   | 24          | [27]           | 0.42        | [4]          | 15.0  | [2] | 23.5   | 15   | MF                          | U                           | C                               | 28.80         | 561.60                | <3 ACTH RIA [27]        |
| <i>Gopherus agassizii</i>                      | 2784        | [4]            |             |              | 100.0 | [2] | -      | 11   | MF                          | U                           | C                               | 5.60          | 20.72                 | ACTH RIA [28]           |
| <i>Heloderma horridum</i>                      | 586         | [29]           |             |              | 30.0  | 7   | M      | U    | C                           | 4.06                        | 7.85                            | <3            | Restraint             | ELISA [29]              |
| <i>Heloderma suspectum</i>                     | 553         | [4]            | 0.14        | [30]         | 48.0  | [2] | 30.0   | 8    | MF                          | U                           | C                               | 3.25          | 3.10                  | <3 Restraint ELISA [31] |
| <i>Hoplodactylus duvaucelii</i>                | 118         | [4]            |             |              | 50.0  | [2] | 18.3   | 16   | MF                          | NB                          | C                               | 5.11          | -                     | >3 - RIA [32]           |
| <i>Lepidochelys kempii</i> <sup>^</sup>        | 4300        | [33]           |             |              | 50.0  | [2] | 25.2   | 18   | U                           | U                           | C                               | 4.41          | 19.02                 | >3 Other RIA [33]       |
| <i>Liopholis whittii</i>                       |             |                |             |              | -     | 14  | M      | NB   | F                           | 5.30                        | 24.80                           | >3            | Restraint             | RIA [34]                |
| <i>Malayopython reticulatus</i>                | 21750       | [35]           | 0.12        | [36]         | 29.4  | [2] | 28.5   | 11   | MF                          | U                           | F                               | 7.20          | 13.20                 | <3 Other ELISA [35]     |
| <i>Natrix tessellata</i>                       | 303         | [37]           |             |              | 14.0  | [2] | -      | 75   | MF                          | B                           | F                               | 13.30         | 69.78                 | >3 Restraint RIA [37]   |
| <i>Nerodia rhombifer</i>                       | 238         | [7]            | 0.53        | [7]          | 10.0  | [2] | -      | 12   | MF                          | NB                          | F                               | 15.16         | 86.10                 | >3 Restraint ELISA [38] |
| <i>Nerodia sipedon</i>                         | 1306        | [4]            |             |              | 21.0  | [2] | -      | 25   | MF                          | NB                          | F                               | 22.99         | 119.86                | >3 Restraint ELISA [38] |

|                                  |       |      |       |      |       |     |      |     |    |    |       |       |        |       |           |       |      |
|----------------------------------|-------|------|-------|------|-------|-----|------|-----|----|----|-------|-------|--------|-------|-----------|-------|------|
| <i>Phrynocephalus vlangalii</i>  |       |      |       | 6.0  | [2]   | -   | 11   | MF  | U  | C  | 10.02 | 19.03 | <3     | Other | EIA       | [39]  |      |
| <i>Pituophis c. affinis</i>      | 457   | [29] | 0.22  | [40] | 33.8  | [2] | 30.0 | 4   | U  | U  | C     | 18.24 | 93.04  | <3    | Restraint | ELISA | [29] |
| <i>Podarcis siculus</i>          | 9     | [7]  | 0.30  | [7]  |       |     | -    | 20  | M  | NB | F     | 14.06 | 64.46  | >3    | Restraint | RIA   | [41] |
| <i>Python bivittatus</i>         | 19050 | [42] |       |      | 28.3  | [2] |      | U   | U  | B  | F     | 24.16 | 73.20  | >3    | Restraint | EIA   | [42] |
| <i>Python regius</i>             | 1086  | [43] | 0.19  | [7]  | 50.0  | [2] | 29.0 | 4   | U  | U  | C     | 37.60 | 56.20  |       | Restraint | RIA   | [43] |
| <i>Regina septemvittata</i>      | 298   | [4]  |       |      | 19.3  | [2] | -    | 15  | MF | NB | F     | 43.05 | 204.50 | >3    | Restraint | ELISA | [38] |
| <i>Sceloporus occidentalis</i>   | 10    | [40] | 27.36 | [40] | 7.0   | [2] | 30.0 | 13  | U  | U  | C     | 29.53 | -      | <3    | -         | RIA   | [44] |
| <i>Sceloporus undulatus</i>      | 6     | [45] | 1.88  | [7]  | 4.0   | [2] | -    | 41  | MF | U  | C     | 21.52 | 48.78  | <3    | Restraint | EIA   | [45] |
| <i>Sphenodon punctatus</i>       | 645   | [46] | 0.17  | [7]  | 100.0 | [2] | -    | 5   | M  | U  | F     | 18.78 | -      |       | -         | RIA   | [46] |
| <i>Thamnophis errans</i>         | 400   | [4]  |       |      | 17.8  | [2] | -    | 51  | M  | U  | F     | 18.89 | 108.67 | <3    | ACTH      | RIA   | [47] |
| <i>Thamnophis s. parietalis</i>  | 37    | [48] |       |      | 20.0  | [2] | -    | 10  | M  | U  | F     | 62.50 | -      |       | -         | RIA   | [48] |
| <i>Tiliqua scincoides</i>        | 303   | [43] | 0.50  | [7]  | 30.0  | [2] | 29.0 | 3   | M  | U  | C     | 8.50  | 9.71   |       | Restraint | RIA   | [43] |
| <i>Urosaurus ornatus</i>         | 4     | [49] |       |      | 3.0   | [2] | -    | 41  | MF | U  | F     | 6.61  | 18.81  | >3    | Restraint | RIA   | [50] |
| <i>Uta stansburiana</i>          | 4     | [40] | 15.96 | [40] | 4.8   | [2] | -    | 173 | F  | U  | F     | 30.20 | 56.50  | <3    | Restraint | RIA   | [51] |
| <i>Varanus salvator</i>          | 7400  | [35] | 0.24  | [52] | 15.8  | [2] | 28.5 | 10  | MF | B  | F     | 3.10  | 11.70  | <3    | Other     | ELISA | [35] |
| <i>Varanus varius</i>            | 4300  | [4]  | 0.28  | [7]  | 22.0  | [2] | -    | 12  | U  | U  | F     | 9.59  | 22.87  | >3    | Restraint | RIA   | [53] |
| <i>Vipera aspis</i>              | 106   | [54] |       |      | 25.0  | [2] | 24.0 | 33  | MF | NB | C     | 11.35 | 34.70  | >3    | Other     | RIA   | [54] |
| <i>Zootoca vivipara</i>          | 4     | [55] | 1.40  | [40] | 12.0  | [2] | 24.5 | 17  | F  | NB | C     | 14.40 | 50.80  | >3    | Restraint | RIA   | [55] |
| Birds                            |       |      |       |      |       |     |      |     |    |    |       |       |        |       |           |       |      |
| <i>Acanthis flammea</i>          | 13    | [56] | 20.63 | [4]  | 12.2  | [4] |      | 78  | MF | B  | F     | 10.50 | 27.50  | <3    | Restraint | RIA   | [57] |
| <i>Aethia pusilla</i>            | 85    | [58] |       |      | 8.1   | [4] |      | 161 | U  | B  | F     | 6.50  | 37.00  | <3    | Restraint | RIA   | [58] |
| <i>Aimophila carpalis</i>        | 16    | [59] |       |      |       |     |      | 41  | M  | B  | F     | 14.00 | 43.03  | <3    | Restraint | ELISA | [59] |
| <i>Amazona ventralis</i> *       | 239   | [60] |       |      |       |     |      | 40  | U  | U  | C     | 19.80 | -      | <3    | -         | RIA   | [60] |
| <i>Anas platyrhynchos</i> ^      | 2709  | [61] | 3.99  | [4]  | 29.1  | [4] |      | 33  | MF | U  | C     | 11.80 | -      | <3    | -         | RIA   | [61] |
| <i>Antigone canadensis</i> *     | 5000  | [62] | 2.07  | [63] | 36.6  | [4] |      | 8   | MF | U  | F     | 1.90  | 25.20  |       | ACTH      | RIA   | [62] |
| <i>Aphelocoma coerulescens</i>   | 76    | [64] |       |      | 15.0  | [4] |      | 35  | MF | U  | F     | 3.00  | 26.10  | <3    | Restraint | RIA   | [64] |
| <i>Aphrastura spinicauda</i>     | 12    | [65] |       |      |       |     |      | 155 | MF | B  | F     | 8.00  | 32.74  | <3    | Restraint | RIA   | [65] |
| <i>Asio otus</i>                 | 262   | [56] | 3.79  | [4]  | 27.8  | [4] |      | 24  | MF | NB | F     | 7.30  | 40.40  | <3    | Restraint | RIA   | [66] |
| <i>Calcarius lapponicus</i>      | 27    | [56] |       |      | 6.0   | [4] |      | 12  | MF | B  | F     | 11.00 | 63.24  | <3    | Restraint | RIA   | [67] |
| <i>Calcarius mccownii</i>        | 25    | [68] |       |      |       |     |      | 33  | MF | B  | C     | 5.90  | 22.00  | <3    | Restraint | RIA   | [69] |
| <i>Calcarius ornatus</i>         | 19    | [56] |       |      |       |     |      | 45  | MF | B  | F     | 5.50  | 21.00  | <3    | Restraint | RIA   | [69] |
| <i>Calcarius pictus</i>          | 27    | [4]  |       |      | 5.0   | [4] |      | 163 | MF | B  | F     | 8.20  | 42.20  | <3    | Restraint | RIA   | [70] |
| <i>Calidris canutus</i>          | 135   | [71] |       |      | 27.3  | [4] |      | U   | U  | U  | F     | 10.00 | -      | >3    | -         | RIA   | [71] |
| <i>Calidris melanotos</i>        | 81    | [68] |       |      |       |     |      | 17  | MF | B  | F     | 10.10 | 51.82  | <3    | Restraint | RIA   | [72] |
| <i>Calidris pusilla</i>          | 26    | [68] |       |      | 19.3  | [4] |      | 52  | MF | B  | F     | 21.60 | 75.46  | <3    | Restraint | RIA   | [72] |
| <i>Columba livia</i>             | 300   | [73] | 5.33  | [63] | 12.6  | [4] |      | 15  | U  | B  | F     | 2.90  | 12.07  | <3    | Restraint | RIA   | [73] |
| <i>Cuculus canorus</i>           | 102   | [74] | 8.22  | [4]  | 12.9  | [4] |      | 25  | MF | B  | F     | 6.10  | -      | >3    | -         | EIA   | [74] |
| <i>Cuculus poliocephalus</i>     | 56    | [74] |       |      |       |     |      | 21  | MF | B  | F     | 6.80  | -      | >3    | -         | EIA   | [74] |
| <i>Cyanistes caeruleus</i>       | 11    | [68] |       |      | 14.6  | [4] |      | U   | MF | B  | F     | 7.20  | 21.20  | <3    | Restraint | RIA   | [68] |
| <i>Daption capense</i>           | 472   | [75] |       |      | 25.0  | [4] |      | 27  | MF | B  | F     | 1.50  | 46.20  |       | Restraint | RIA   | [75] |
| <i>Diomedea exulans</i>          | 7650  | [56] | 2.26  | [7]  | 50.0  | [4] |      | 75  | MF | B  | F     | 4.35  | -      |       | -         | RIA   | [75] |
| <i>Empidonax oberholseri</i>     | 21    | [68] |       |      | 9.0   | [4] |      | U   | MF | B  | F     | 29.10 | 43.45  | <3    | Restraint | RIA   | [68] |
| <i>Eudyptes chrysocome</i>       | 2500  | [56] |       |      |       |     |      | 7   | U  | B  | F     | 5.80  | 50.19  | <3    | Restraint | RIA   | [76] |
| <i>Ficedula hypoleuca</i>        | 12    | [77] | 19.33 | [4]  | 15.0  | [4] |      | U   | MF | B  | F     | 24.90 | 54.52  | <3    | Other     | RIA   | [77] |
| <i>Gallus gallus</i>             | 2580  | [4]  | 2.22  | [7]  |       |     |      | 6   | F  | U  | C     | 1.20  | 25.26  |       | ACTH      | EIA   | [78] |
| <i>Haemorhous mexicanus</i>      | 23    | [79] | 13.64 | [4]  | 11.6  | [4] |      | 6   | M  | B  | F     | 5.90  | -      | >3    | -         | RIA   | [79] |
| <i>Histrionicus histrionicus</i> | 623   | [56] |       |      | 14.6  | [4] |      | 38  | MF | B  | F     | 8.30  | 34.40  | <3    | Restraint | RIA   | [80] |
| <i>Horornis diphone</i>          | 17    | [68] |       |      |       |     |      | U   | M  | B  | F     | 52.20 | 86.90  | <3    | Restraint | RIA   | [68] |
| <i>Junco hyemalis</i>            | 20    | [56] | 16.40 | [4]  | 11.3  | [4] |      | 40  | MF | B  | F     | 15.30 | 50.71  | <3    | Restraint | RIA   | [81] |
| <i>Kieneria aberti</i>           | 46    | [56] | 15.76 | [4]  | 8.6   | [4] |      | 31  | M  | B  | F     | 10.00 | 42.91  | <3    | Restraint | RIA   | [82] |

|                                  |      |          |       |       |      |     |     |    |    |   |       |       |    |           |       |            |
|----------------------------------|------|----------|-------|-------|------|-----|-----|----|----|---|-------|-------|----|-----------|-------|------------|
| <i>Lagopus muta</i>              | 422  | [56]     |       |       |      |     | 9   | MF | U  | F | 16.50 | 52.89 | <3 | Restraint | RIA   | [83]       |
| <i>Larus hyperboreus</i>         | 1576 | [75]     | 7.21  | [63]  | 21.1 | [4] | 38  | MF | B  | F | 12.20 | 33.95 |    | Restraint | RIA   | [75]       |
| <i>Leuconotopicus borealis</i>   | 44   | [56]     |       |       |      |     | 113 | MF | B  | F | 10.00 | 30.42 | <3 | Restraint | RIA   | [84]       |
| <i>Limosa lapponica</i>          | 216  | [85]     |       |       | 34.0 | [4] | U   | MF | NB | F | 13.00 | 65.30 | >3 | Restraint | RIA   | [85]       |
| <i>Megadyptes antipodes</i>      | 5200 | [68]     | 2.40  | [86]  |      |     | 16  | MF | B  | F | 0.50  | 17.30 | >3 | Restraint | EIA   | [87]       |
| <i>Melanitta deglandi</i>        | 1194 | [88]     |       |       |      |     | 12  | MF | U  | C | 4.00  | 43.75 | <3 | ACTH      | EIA   | [88]       |
| <i>Melospiza melodia</i>         | 21   | [56]     | 13.09 | [4]   | 11.3 | [4] | U   | MF | B  | F | 18.50 | -     | >3 | -         | RIA   | [89]       |
| <i>Mimus polyglottos</i>         | 49   | [68]     |       |       | 20.0 | [4] | U   | MF | B  | F | 4.00  | 29.40 | <3 | Restraint | RIA   | [68]       |
| <i>Morus serrator</i>            | 2350 | [56]     |       |       | 33.0 | [4] | 7   | MF | B  | F | 3.10  | 55.87 | <3 | Restraint | RIA   | [90]       |
| <i>Pagodroma nivea</i>           | 419  | [75]     |       |       | 46.0 | [4] | 35  | MF | B  | F | 4.50  | 43.15 |    | Restraint | RIA   | [75]       |
| <i>Parabuteo unicinctus</i>      | 886  | [91]     |       |       |      |     | U   | U  | U  | F | 8.00  | -     |    | -         | RIA   | [91]       |
| <i>Parus major</i>               | 19   | [68]     | 21.79 | [63]  | 15.4 | [4] | U   | M  | B  | F | 19.90 | -     | <3 | -         | RIA   | [68]       |
| <i>Passer domesticus</i>         | 29   | [68]     | 12.99 | [63]  | 23.0 | [4] | U   | MF | B  | F | 4.50  | 33.20 | <3 | Restraint | RIA   | [68]       |
| <i>Passerculus sandwichensis</i> | 20   | [68]     | 13.90 | [4]   |      |     | U   | MF | B  | F | 20.40 | 67.60 | <3 | Restraint | RIA   | [68]       |
| <i>Passerina cyanea</i>          | 15   | [68]     |       |       | 11.0 | [4] | U   | MF | B  | F | 21.90 | 63.30 | <3 | Restraint | RIA   | [68]       |
| <i>Pardalipicus ashiaticus</i>   | 35   | [92]     |       |       |      |     | 6   | M  | B  | F | 11.00 | 52.05 |    | Restraint | ELISA | [92]       |
| <i>Petrochelidon pyrrhonota</i>  | 23   | [68]     |       |       | 11.8 | [4] | U   | MF | B  | F | 9.00  | -     | <3 | -         | RIA   | [68]       |
| <i>Phaethon rubricauda</i>       | 718  | [68]     | 5.61  | [86]  | 32.7 | [4] | U   | MF | B  | F | 5.20  | 33.50 | <3 | Restraint | RIA   | [68]       |
| <i>Phalaropus fulicarius</i>     | 56   | [68]     |       |       |      |     | U   | MF | B  | F | 21.80 | 75.40 | <3 | Restraint | RIA   | [68]       |
| <i>Plectrophenax nivalis</i>     | 34   | [68]     | 13.09 | [63]  | 9.6  | [4] | U   | MF | B  | F | 9.30  | 26.15 | <3 | Restraint | RIA   | [68]       |
| <i>Pterodroma macroptera</i>     | 570  | [68]     |       |       |      |     | U   | MF | B  | F | 12.60 | 67.20 | <3 | Restraint | RIA   | [68]       |
| <i>Ptilotula penicillata</i>     | 19   | [68]     |       |       |      |     | 9   | M  | U  | F | 15.00 | 49.48 | <3 | Restraint | RIA   | [93]       |
| <i>Pygoscelis adeliae</i>        | 4850 | [56]     | 3.08  | [7]   |      |     | 13  | MF | B  | F | 7.80  | -     | <3 | -         | RIA   | [94]       |
| <i>Rissa brevirostris</i>        | 377  | [68]     |       |       | 27.0 | [4] | U   | M  | B  | F | 2.70  | 32.50 | <3 | Restraint | RIA   | [68]       |
| <i>Rissa tridactyla</i>          | 411  | [95]     |       |       | 28.5 | [4] | U   | M  | B  | F | 7.30  | 45.22 | <3 | Other     | RIA   | [95]       |
| <i>Saxicola torquata</i>         | 15   | [68]     |       |       | 8.8  | [4] | U   | MF | B  | F | 10.10 | 35.50 | <3 | Restraint | RIA   | [68]       |
| <i>Setophaga petechia</i>        | 10   | [68]     |       |       | 11.0 | [4] | U   | M  | B  | F | 22.30 | 86.10 | <3 | Restraint | RIA   | [68]       |
| <i>Setophaga ruticilla</i>       | 8    | [56]     |       |       | 10.1 | [4] | 165 | MF | NB | F | 15.00 | 32.18 | <3 | Restraint | RIA   | [96]       |
| <i>Somateria mollissima</i>      | 1140 | [75]     |       |       | 37.8 | [4] | 55  | F  | B  | F | 6.00  | 40.40 |    | Restraint | RIA   | [75]       |
| <i>Spheniscus humboldti</i>      | 5000 | [56]     | 2.45  | [7]   |      |     | 17  | U  | B  | F | 3.50  | 25.78 | <3 | Restraint | RIA   | [76]       |
| <i>Spheniscus magellanicus</i>   | 4500 | [56]     |       |       | 30.0 | [4] | 8   | U  | B  | F | 4.20  | 34.68 | <3 | Restraint | RIA   | [76]       |
| <i>Spheniscus mendiculus</i>     | 2500 | [56]     |       |       |      |     | 9   | U  | B  | F | 3.00  | 12.09 | <3 | Restraint | RIA   | [76]       |
| <i>Spinus pinus</i>              | 17   | [97]     | 21.45 | [4]   | 11.0 | [4] | U   | MF | U  | F | 8.50  | 19.68 | <3 | Other     | RIA   | [97]       |
| <i>Spizelloides arborea</i>      | 18   | [68]     | 19.66 | [63]  | 10.8 | [4] | U   | MF | B  | F | 16.40 | 37.35 | <3 | Restraint | RIA   | [68]       |
| <i>Stercorarius maccormicki</i>  | 1419 | [75]     |       |       | 23.5 | [4] | 31  | MF | B  | F | 6.70  | -     |    | -         | RIA   | [75]       |
| <i>Sterna hirundo</i>            | 125  | [68]     |       |       | 33.0 | [4] | U   | MF | B  | F | 10.00 | 31.60 | <3 | Restraint | RIA   | [68]       |
| <i>Sturnus vulgaris</i>          | 74   | [56]     | 11.69 | [4]   | 22.9 | [4] | U   | MF | B  | F | 9.90  | 33.37 | <3 | Restraint | EIA   | [98]       |
| <i>Sula granti</i>               | 1754 | [4]      |       | [4]   | 21.0 | [4] | 79  | U  | NB | F | 2.10  | -     | <3 | -         | RIA   | [99]       |
| <i>Sula nebulosa</i>             | 1542 | [68]     |       |       | 18.0 | [4] | U   | MF | B  | F | 16.10 | 37.70 | <3 | Restraint | RIA   | [68]       |
| <i>Sula sula</i>                 | 850  | [100]    | 4.28  | [86]  | 23.0 | [4] | 10  | MF | NB | F | 6.20  | -     | >3 | -         | RIA   | [100]      |
| <i>Tachycineta bicolor</i>       | 20   | [56]     |       |       | 12.1 | [4] | 9   | F  | B  | F | 8.20  | -     | >3 | -         | EIA   | [101]      |
| <i>Taeniopygia guttata</i>       | 12   | [68]     |       |       | 12.0 | [4] | U   | MF | B  | F | 4.30  | 22.20 | <3 | Restraint | RIA   | [68]       |
| <i>Thalassarche melanophrys</i>  | 3552 | [56,102] | 4.47  | [103] | 43.7 | [4] | 69  | MF | B  | F | 2.30  | -     | <3 | -         | RIA   | [102, 104] |
| <i>Thryophilus rufalbus</i>      | 27   | [56]     |       |       |      |     | 54  | MF | B  | F | 9.00  | 37.83 | <3 | Restraint | EIA   | [105]      |
| <i>Toxostoma curvirostre</i>     | 79   | [56]     |       |       | 10.8 | [4] | 26  | M  | B  | F | 15.00 | 39.90 | <3 | Restraint | RIA   | [82]       |
| <i>Turdus merula</i>             | 113  | [56]     |       |       | 21.8 | [4] | 112 | MF | NB | F | 4.70  | -     | <3 | -         | EIA   | [106]      |
| <i>Uria lomvia</i>               | 1098 | [58]     | 6.87  | [4]   | 29.0 | [4] | 141 | U  | B  | F | 3.80  | 46.41 | <3 | Restraint | RIA   | [58]       |
| <i>Zenaidura macroura</i>        | 119  | [56]     | 5.98  | [4]   | 31.3 | [4] | 65  | MF | B  | F | 3.80  | 29.59 | <3 | Restraint | RIA   | [107]      |
| <i>Zonotrichia capensis</i>      | 20   | [68]     |       |       |      |     | U   | MF | B  | F | 24.20 | 51.40 | <3 | Restraint | RIA   | [68]       |

|                                    |         |       |       |       |      |     |    |    |    |   |        |         |         |           |           |       |       |
|------------------------------------|---------|-------|-------|-------|------|-----|----|----|----|---|--------|---------|---------|-----------|-----------|-------|-------|
| <i>Zonotrichia leucophrys</i>      | 25      | [68]  | 12.87 | [4]   | 13.3 | [4] | U  | MF | B  | F | 19.30  | 64.90   | <3      | Restraint | RIA       | [68]  |       |
| Mammals, cortisol dominant         |         |       |       |       |      |     |    |    |    |   |        |         |         |           |           |       |       |
| <i>Aotus trivirgatus</i>           | 800     | [4]   | 2.73  | [4]   | 30.1 | [4] | 7  | U  | U  | C | 12.00  | -       | 300.00  | -         | RIA       | [108] |       |
| <i>Bos taurus</i>                  | 347000  | [7]   | 0.88  | [4]   | 20.0 | [4] | 7  | U  | U  | C | 0.70   | 7.60    | 3.10    | ACTH      | RIA       | [109] |       |
| <i>Callospermophilus saturatus</i> | 275     | [4]   | 3.31  | [4]   |      |     | 54 | MF | U  | F | 9.60   | -       | 16.80   | -         | RIA       | [110] |       |
| <i>Canis l. dingo</i>              | 26625   | [4]   |       |       | 20.6 | [4] | 6  | MF | NB | C | 3.30   | 16.00   | 19.00   | <3        | ACTH      | DIDA  | [111] |
| <i>Castor fiber</i>                | 19100   | [112] |       |       | 20.6 | [4] | 34 | MF | B  | F | 9.60   | -       | 22.20   | >3        | -         | RIA   | [112] |
| <i>Cavia porcellus</i>             | 728     | [4]   | 3.33  | [4]   | 12.0 | [4] | U  | M  | U  | C | 19.40  | -       | 190.30  | <3        | -         | RIA   | [113] |
| <i>Ctenomys talarum</i>            | 170     | [114] | 5.05  | [4]   |      |     | 6  | M  | B  | F | 2.30   | 3.13    | 20.10   | <3        | Restraint | RIA   | [114] |
| <i>Dasyurus geoffroii</i>          | 890     | [4]   | 2.25  | [4]   | 6.2  | [4] | 4  | MF | NB | C | 1.50   | -       | 18.00   | <3        | -         | DIDA  | [111] |
| <i>Dasyurus maculatus</i>          | 5500    | [4]   | 1.76  | [4]   | 6.8  | [4] | 1  | U  | NB | C | 3.90   | -       | 29.00   | <3        | -         | DIDA  | [111] |
| <i>Elephas maximus*</i>            | 3672000 | [4]   | 0.64  | [4]   | 79.6 | [4] | 5  | F  | B  | C | 4.80   | -       | 9.10    | -         | EIA       | [115] |       |
| <i>Macaca fascicularis</i>         | 6363    | [4]   | 2.73  | [116] | 39.0 | [4] | 9  | U  | U  | C | 13.80  | -       | 210.00  | -         | RIA       | [108] |       |
| <i>Macaca mulatta</i>              | 8235    | [4]   | 2.43  | [116] | 40.0 | [4] | 9  | U  | U  | C | 8.80   | -       | 245.00  | -         | RIA       | [108] |       |
| <i>Macropus giganteus</i>          | 49500   | [4]   |       |       | 25.0 | [4] | 7  | MF | NB | C | 2.20   | -       | 18.00   | <3        | -         | DIDA  | [111] |
| <i>Mustela p. furo</i>             | 1140    | [117] |       |       | 11.1 | [4] | 8  | MF | U  | C | 2.50   | 25.80   | 26.60   | ACTH      | RIA       | [117] |       |
| <i>Notamacropus rufogriseus</i>    | 16200   | [4]   |       |       | 15.2 | [4] | U  | U  | U  | C | 1.50   | -       | 17.00   | <3        | -         | DIDA  | [118] |
| <i>Odocoileus virginianus</i>      | 87000   | [4]   | 1.99  | [4]   | 23.0 | [4] | U  | U  | U  | F | 2.60   | -       | 56.00   | -         | RIA       | [119] |       |
| <i>Orcinus orca</i>                | 3987500 | [4]   | 0.57  | [120] |      |     | 14 | M  | U  | C | 2.80   | -       | 18.90   | -         | EIA       | [121] |       |
| <i>Oryctolagus cuniculus</i>       | 3000    | [122] | 3.41  | [4]   | 9.0  | [4] | U  | U  | U  | C | 11.00  | -       | 31.50   | -         | EIA       | [122] |       |
| <i>Ovis aries</i>                  | 80000   | [4]   |       |       | 22.8 | [4] | U  | U  | NB | C | 0.90   | -       | 5.20    | <3        | -         | DIDA  | [111] |
| <i>Psammomys obesus</i>            | 212     | [4]   |       |       | 3.2  | [4] | U  | M  | U  | F | 5.30   | -       | 166.50  | <3        | -         | RIA   | [123] |
| <i>Saimiri sciureus</i>            | 925     | [4]   | 5.29  | [4]   | 30.2 | [4] | 14 | U  | U  | C | 20.50  | -       | 1990.00 | -         | RIA       | [108] |       |
| <i>Setonix brachyurus</i>          | 3500    | [4]   | 1.74  | [4]   | 13.8 | [4] | 7  | MF | NB | C | 5.90   | -       | 12.00   | <3        | -         | DIDA  | [111] |
| <i>Sus scrofa</i>                  | 105000  | [124] | 0.77  | [4]   | 27.0 | [4] | 10 | U  | B  | C | 11.10  | 12.05   | 17.30   | Other     | RIA       | [124] |       |
| <i>Tamias amoenus</i>              | 48      | [4]   | 9.42  | [7]   |      |     | 20 | F  | B  | F | 30.2   | 59.03   | 658.29  | <3        | Restraint | RIA   | [125] |
| <i>Urocitellus beldingi*</i>       | 290     | [4]   | 2.71  | [4]   |      |     | 14 | MF | U  | F | 16.30  | -       | 37.70   | <3        | -         | RIA   | [126] |
| <i>Urocitellus parryii</i>         | 650     | [7]   | 4.47  | [7]   |      |     | 27 | M  | U  | F | 8.10   | -       | 66.20   | >3        | -         | RIA   | [127] |
| <i>Vombatus ursinus</i>            | 25500   | [4]   |       |       | 30.0 | [4] | U  | U  | U  | C | 1.30   | -       | 2.30    | <3        | -         | DIDA  | [118] |
| <i>Vulpes vulpes</i>               | 4132    | [4]   | 3.00  | [4]   | 21.3 | [4] | U  | U  | NB | C | 5.90   | -       | 23.00   | <3        | -         | DIDA  | [111] |
| <i>Wallabia bicolor</i>            | 14625   | [4]   |       |       | 16.8 | [4] | 3  | MF | NB | C | 1.20   | -       | 11.00   | <3        | -         | DIDA  | [111] |
| Mammals, CCST dominant             |         |       |       |       |      |     |    |    |    |   |        |         |         |           |           |       |       |
| <i>Mesocricetus auratus</i>        | 105     | [4]   | 8.40  | [7]   | 3.9  | [4] | U  | M  | U  | C | 80.00  | 284.02  | 2.50    | Other     | EIA       | [128] |       |
| <i>Microtus ochrogaster</i>        | 40      | [4]   | 8.51  | [7]   | 5.3  | [4] | U  | U  | U  | C | 631.00 | -       |         | <3        | -         | RIA   | [129] |
| <i>Microtus pennsylvanicus</i>     | 49      | [4]   | 10.75 | [7]   |      |     | 77 | MF | U  | F | 390.30 | -       |         | <3        | -         | RIA   | [130] |
| <i>Mus musculus*</i>               | 33      | [131] | 8.20  | [131] | 4.0  | [4] | 6  | F  | B  | C | 93.6   | 1183.00 | 12.72   | Restraint | RIA       | [132] |       |
| <i>Tachyglossus aculeatus</i>      | 3500    | [4]   | 0.88  | [7]   | 49.5 | [4] | 1  | F  | NB | C | 3.50   | -       | 1.80    | <3        | -         | DIDA  | [111] |

<sup>1</sup>Stressors categorized as “Other” include synthetic angiotensin I injection, oxytocin injection, glucocorticoid implant, fasting, transport, extreme temperature, and carbon dioxide exposure.

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515



**Table S2:** Results for model selection using Bayesian Information Criterion (BIC). Model groupings are categorized by numbers 1-3 in the first column. A series of additive models were evaluated to assess the possible effects of a range of covariates on model fits. Covariates considered here included “Vert” (reptile, bird, and mammal classes), “Assay” (RIA, EIA, ELISA, or DIDA), “Time” (time to bleed at <3 min. or >3 min.), “Envr” (captive or field), “Season” (breeding or non-breeding individuals), “Sex” (males, females, or both), and “Stressor” (ACTH injections, restraint, or other). Bolded delta BIC values indicate the simplest model as the most parsimonious or having minimal difference from the most parsimonious model ( $\Delta\text{BIC} < 2.3$ ).

|                                            | df | BIC    | $\Delta\text{BIC}$ |
|--------------------------------------------|----|--------|--------------------|
| <b>Baseline CCST vs. Body Mass</b>         |    |        |                    |
| 1: Baseline CCST vs. Body Mass             | 4  | 426.61 | <b>0</b>           |
| 1: Baseline CCST vs. Body Mass + Vert      | 6  | 433.45 | 6.84               |
| 2: Baseline CCST vs. Body Mass             | 4  | 334.49 | <b>0</b>           |
| 2: Baseline CCST vs. Body Mass + Assay     | 5  | 340.11 | 5.62               |
| 2: Baseline CCST vs. Body Mass + Time      | 7  | 341.53 | 7.04               |
| 3: Baseline CCST vs. Body Mass             | 4  | 210.81 | <b>0</b>           |
| 3: Baseline CCST vs. Body Mass + Envr      | 6  | 213.70 | 2.89               |
| 3: Baseline CCST vs. Body Mass + Season    | 5  | 215.74 | 4.93               |
| 3: Baseline CCST vs. Body Mass + Sex       | 6  | 218.55 | 7.74               |
| <b>Baseline CCST vs. MSMR</b>              |    |        |                    |
| 1: Baseline CCST vs. MSMR                  | 4  | 242.16 | <b>0</b>           |
| 1: Baseline CCST vs. MSMR + Vert           | 6  | 243.06 | 0.90               |
| 2: Baseline CCST vs. MSMR                  | 4  | 177.30 | <b>0</b>           |
| 2: Baseline CCST vs. MSMR + Assay          | 7  | 179.53 | 2.23               |
| 2: Baseline CCST vs. MSMR + Time           | 5  | 181.53 | 4.23               |
| 3: Baseline CCST vs. MSMR + Envr           | 6  | 103.62 | <b>0</b>           |
| 3: Baseline CCST vs. MSMR                  | 4  | 105.91 | <b>2.29</b>        |
| 3: Baseline CCST vs. MSMR + Sex            | 6  | 107.96 | 4.34               |
| 3: Baseline CCST vs. MSMR + Season         | 5  | 108.94 | 5.32               |
| <b>Elevated CCST vs. Baseline CCST</b>     |    |        |                    |
| 1: Elevated CCST vs. Baseline CCST         | 4  | 220.07 | <b>0</b>           |
| 1: Elevated CCST vs. Baseline CCST + Vert  | 6  | 228.72 | 8.65               |
| 2: Elevated CCST vs. Baseline CCST         | 4  | 165.15 | <b>0</b>           |
| 2: Elevated CCST vs. Baseline CCST + Time  | 5  | 170.16 | 5.01               |
| 2: Elevated CCST vs. Baseline CCST + Assay | 7  | 176.68 | 11.53              |

|                                               |   |        |          |
|-----------------------------------------------|---|--------|----------|
| 3: Elevated CCST vs. Baseline CCST            | 4 | 115.61 | <b>0</b> |
| 3: Elevated CCST vs. Baseline CCST + Envr     | 6 | 119.24 | 3.63     |
| 3: Elevated CCST vs. Baseline CCST + Season   | 5 | 121.03 | 5.42     |
| 3: Elevated CCST vs. Baseline CCST + Stressor | 6 | 122.40 | 6.79     |
| 3: Elevated CCST vs. Baseline CCST + Sex      | 6 | 124.76 | 9.15     |

**Figure S2:** Plots to evaluate effects of corticosterone on lifespan for reptiles, birds, and mammals. Plots across the top row (Panels A-C) show lifespan (years) versus baseline corticosterone (CCST) concentrations (ng/ml). Plots across the bottom row (Panels D-F) show the residuals of a plot of lifespan versus mass-specific metabolic rate (MSMR; W/g) plotted against baseline corticosterone concentrations. Data are natural log-transformed, and lines are fit using ordinary least-squares (OLS) linear regression. Shading indicates 95% confidence intervals.

